NYC Parking Tickets Violation Analytics

Data Visualisation & Analytics Project Report

Sector: Urban Governance, Public Policy, Transportation Analytics, Municipal Revenue Intelligence
Project Theme: NYC Parking Tickets — Enforcement, Revenue, Safety & Repeat Offender Analysis
Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, SciPy, Statsmodels, Tableau, Jupyter Notebook
Dashboard Platform: Tableau Public / Tableau Desktop
Dataset: NYC Parking Tickets Sample Dataset
GitHub Repository: GitHub - KatalKavya96/SectionC_G12_NYC_Ticketing_Analysis

EXECUTIVE SUMMARY

NYC Parking Enforcement Strategy and City-Level Decision Intelligence

THE PROBLEM

New York City issues a large number of parking tickets every year, but the real challenge is not only the volume of violations. The deeper issue is understanding **where violations occur, which violations matter most, which boroughs require different enforcement strategies, and whether enforcement is improving safety or mainly generating revenue**.

A uniform enforcement strategy may appear simple, but the data suggests that violation behavior differs across boroughs, vehicle types, streets, offender groups, and time periods. Some violations are avoidable and can be reduced through better signage, while others are safety-critical and require stricter enforcement. Similarly, repeat offenders may be responsible for a disproportionate share of ticket volume and fine revenue.

This project converts raw NYC parking ticket data into a structured decision-support system for city officials, enforcement teams, policy analysts, and urban planners.

THE APPROACH

The project follows a complete analytics pipeline: raw data ingestion, data auditing, cleaning, transformation, exploratory data analysis, pivot-based aggregation, KPI design, statistical testing, and Tableau dashboard development. The workflow is divided into five Jupyter notebooks covering extraction, cleaning, EDA, statistical analysis, and final Tableau export preparation. The blueprint includes a 12-step cleaning pipeline, 10 pivot tables, 10 EDA analyses, 15 KPIs, 12 Tableau sheets, 4 dashboards, 6 statistical tests, and 5 data-backed recommendations.

The final output is a Tableau-based analytical system that helps answer four major questions:

1. How big is the city-wide parking violation problem?
2. Where is fine revenue concentrated?
3. Which violations are safety-critical versus avoidable?
4. Which offender groups and vehicle types require targeted policy action?

KEY INSIGHTS

1. Enforcement Volume Alone Is Not Enough

Total ticket count gives scale, but it does not explain enforcement quality. A borough may issue many low-value administrative tickets, while another may issue fewer but higher-severity safety-critical violations. Therefore, ticket count must be analyzed together with revenue, average fine, violation severity, and safety impact.

2. Boroughs Require Different Enforcement Policies

The Borough × Violation pivot table and normalized heatmap are central to this project because they reveal whether violation patterns differ across boroughs. The project explicitly uses borough-violation cross-tabulation for chi-square testing to validate whether borough-specific enforcement policies are statistically justified.

3. Repeat Offenders Are a High-Impact Policy Segment

The repeat offender Pareto analysis is one of the strongest findings in the project. It shows whether a small percentage of plates generate a large share of fine revenue. This provides a direct basis for escalation programs, DMV flagging, warning letters, and booting eligibility.

4. Avoidable Violations Represent a Prevention Opportunity

Avoidable violations such as street cleaning, expired meter, and time-limit violations indicate where better communication, signage, and digital parking infrastructure could reduce violations. This shifts the city strategy from only penalising drivers to preventing unnecessary violations.

5. Commercial Vehicles Require Infrastructure-Based Solutions

Commercial and taxi vehicles may contribute disproportionately to safety-critical violations such as fire hydrant, bus stop, and blocking intersection violations. If this pattern appears in the dashboard, the solution is not only more tickets but better loading-zone design.

SECTION 3 — SECTOR & BUSINESS CONTEXT

Parking enforcement is a major part of urban governance. In cities like New York, parking tickets affect municipal revenue, road safety, traffic flow, citizen compliance, and public trust. Poorly designed enforcement may generate revenue but fail to improve safety. Similarly, uniform enforcement may ignore local differences between boroughs, streets, vehicle groups, and repeat offenders.

This project is relevant to several stakeholders:

City Transportation Departments: To optimise enforcement deployment.
Parking Enforcement Agencies: To allocate officers by borough, day, and violation type.
Policy Teams: To design escalation programs and safety-first policies.
Urban Planners: To identify streets needing better signage or loading zones.
Finance Departments: To understand revenue leakage and collection risk.
Citizens: To benefit from clearer rules, safer streets, and fairer enforcement.

The problem was selected because parking ticket data is not just administrative data. It contains strong signals about city behavior, transportation stress, infrastructure gaps, commercial vehicle pressure, and compliance patterns.

SECTION 4 — PROBLEM STATEMENT & OBJECTIVES

Problem Statement

To analyze NYC parking ticket violations and identify actionable patterns in enforcement volume, fine revenue, borough-level violation behavior, repeat offenders, vehicle types, avoidable violations, and safety-critical violations, and to convert these findings into Tableau dashboards and data-backed policy recommendations.

Objectives

1. Clean and transform raw parking ticket data into an analysis-ready dataset.
2. Build a structured ETL pipeline with audit logs and validation checks.
3. Create meaningful KPIs for enforcement, revenue, safety, and data quality.
4. Use pivot tables to uncover borough, time, vehicle, street, and offender patterns.
5. Perform EDA to identify trends, outliers, seasonality, and violation concentration.
6. Conduct statistical tests to validate important relationships.
7. Design four Tableau dashboards for different decision-making audiences.
8. Generate data-backed recommendations for enforcement, policy, infrastructure, and revenue recovery.

Scope

The project focuses on NYC parking ticket records and their analytical use for:

- Borough-level comparison
- Fine revenue analysis
- Violation severity analysis
- Repeat offender identification
- Vehicle type profiling
- Street-level hotspot detection
- Out-of-state ticket analysis
- Tableau dashboard storytelling

Out of Scope

- Real-time enforcement prediction
- Individual driver identity analysis
- Legal adjudication outcomes
- Payment collection history unless separately available
- Live integration with DMV or enforcement systems

Success Criteria

The project is successful if it produces:

- A clean and validated dataset
- 15 business-ready KPIs
- 10 pivot-based analytical tables
- 10 EDA visualisations
- 6 statistical tests with plain-English interpretation
- 12 Tableau sheets
- 4 complete dashboards
- 5 actionable, evidence-backed recommendations

SECTION 5 — DATA DESCRIPTION

5.1 Data Source

The dataset used in this project is an NYC parking tickets sample dataset stored locally in the project directory.

Local File: data/raw/nyc_parking_tickets_sample.csv
Domain: Urban parking enforcement and municipal violation analytics
Primary Unit of Analysis: One parking ticket record
Main Identifier: Summons_Number

5.2 Dataset Coverage

The dataset includes parking ticket-level information such as:

- Summons number
- Issue date
- Violation code
- Violation description
- Violation county / borough

- Fine amount
- Plate ID
- Registration state
- Vehicle make
- Vehicle body type
- Vehicle color
- Street name
- Vehicle year

After cleaning, the project expects a processed dataset named nyc_parking_clean.csv with 38 columns and 0 null values, as defined in the project blueprint.

5.3 Key Original Columns

Column	Description
Summons_Number	Unique ticket/summons identifier
Issue_Date	Date on which ticket was issued
Violation_Code	Numeric or categorical violation code
Violation_Description	Description of parking violation
Violation_County	Borough/county where violation occurred
Fine_Amount	Fine amount associated with ticket
Plate_ID	Vehicle plate identifier
Registration_State	Vehicle registration state
Vehicle_Make	Vehicle manufacturer
Vehicle_Body_Type	Vehicle body type
Vehicle_Color	Vehicle color
Street_Name	Street where ticket was issued

5.4 Engineered Features

Feature	Description
Month_Name	Month extracted from issue date
Month	Numeric month
Day_of_Week	Day name extracted from issue date
Day_Num	Numeric day of week
Is_Weekend	Weekend flag
Is_OutOfState	Whether registration state is outside NY
State_Group	NY / NJ / CT / Other / Unknown grouping
Vehicle_Age	Age of vehicle based on vehicle year
Vehicle_Age_Group	Vehicle age bucket
Vehicle_Type_Group	Passenger, Commercial, Taxi, Transit, Other
Violation_Severity	Safety-Critical, Standard, Administrative
Is_Avoidable	Flag for preventable violations
Ticket_Count_Per_Plate	Number of tickets per plate
Is_Repeat_Offender	Repeat offender flag
Offender_Tier	1, 2–3, 4–5, 6+ ticket tier
Fine_Category	Fine amount category
Is_Complete_Record	Data quality completeness flag

5.5 Data Quality Issues

The raw dataset required cleaning due to:

- Multiple date formats
- Missing or invalid fine values
- Invalid vehicle years such as 9999
- Inconsistent vehicle color abbreviations
- Mixed-case borough names
- Inconsistent vehicle make spellings
- Invalid registration state codes
- Duplicated summons numbers
- Street name formatting issues

SECTION 6 — DATA CLEANING & TRANSFORMATION

6.1 ETL Pipeline Overview

The ETL pipeline was designed as a structured five-notebook workflow:

1. 01_extraction.ipynb — Raw ingestion and audit
2. 02_cleaning.ipynb — 12-step cleaning pipeline
3. 03_eda.ipynb — Exploratory data analysis
4. 04_statistical_analysis.ipynb — Hypothesis testing
5. 05_final_load_prep.ipynb — KPI computation and Tableau exports

The extraction notebook loads the raw CSV as strings, audits shape, dtypes, null percentages, unique counts, and sample values. The cleaning notebook applies a 12-step transformation process and outputs a clean dataset and transformation log.

6.2 Major Cleaning Steps

Step	Cleaning Action	Purpose
1	Multi-format date parsing	Standardise all dates into one format
2	Fine amount correction	Remove invalid zero/negative fines and impute where required
3	Vehicle year validation	Remove impossible or future year values
4	Color mapping	Convert abbreviations like GY, BK, WH into full names
5	Borough standardisation	Standardise borough names and abbreviations
6	Vehicle make consolidation	Merge inconsistent make names
7	Body type grouping	Group body codes into Passenger, Commercial, Taxi, Transit
8	State validation	Identify invalid states and create out-of-state flag
9	Street cleaning	Normalize street names and extract street type
10	Plate repeat detection	Create repeat offender metrics
11	Duplicate removal	Remove duplicate summons numbers
12	Validation assertions	Confirm final dataset consistency

6.3 Key Transformations

The project creates policy-focused analytical features, including:

- Is_OutOfState for revenue leakage analysis
- Is_Repeat_Offender for offender escalation analysis
- Violation_Severity for safety prioritization
- Is_Avoidable for prevention-focused recommendations
- Vehicle_Type_Group for commercial and passenger analysis
- Offender_Tier for Pareto and repeat offender dashboards
- Is_Complete_Record for data governance KPI

6.4 Output Dataset

Output File: data/processed/nyc_parking_clean.csv
Expected Output: 38 columns, 0 nulls
Additional Output: docs/etl_transformation_log.json

The transformation log documents each cleaning step and supports the ETL methodology section of the report.

SECTION 7 — KPI & METRIC FRAMEWORK

The project defines 15 KPIs for Tableau dashboards. These KPIs are designed to support operational, financial, safety, compliance, and data governance decisions.

KPI	Formula	Business Meaning
Total Tickets Issued	COUNT([Summons_Number])	Measures total enforcement volume
Total Fine Revenue	SUM([Fine_Amount])	Measures total financial value
Avg Fine per Ticket	SUM([Fine_Amount]) / COUNT([Summons_Number])	Shows enforcement quality and severity
Out-of-State Ticket %	SUM([Is_OutOfState]) / COUNT(*) * 100	Indicates collection risk and revenue leakage
Repeat Offender Rate	SUM([Is_Repeat_Offender]) / COUNT(*) * 100	Measures habitual violation behavior
Safety Violation %	COUNT(Safety-Critical) / COUNT(*) * 100	Measures public safety focus
Avoidable Violation %	SUM([Is_Avoidable]) / COUNT(*) * 100	Shows prevention opportunity
Unique Vehicles Ticketed	COUNTD([Plate_ID])	Measures enforcement breadth
Commercial Vehicle %	COUNT(COM plates) / COUNT(*) * 100	Supports loading zone planning
Peak Month Ticket Count	MAX(monthly ticket count)	Supports staffing capacity planning
Top Revenue Borough	MAX(SUM([Fine_Amount]) by Borough)	Identifies priority enforcement zone
Revenue per Unique Vehicle	SUM([Fine_Amount]) / COUNTD([Plate_ID])	Shows enforcement concentration
Weekend Enforcement %	SUM([Is_Weekend]) / COUNT(*) * 100	Measures weekend deployment intensity
Top Street Revenue	MAX(SUM([Fine_Amount]) by Street)	Identifies highest-yield hotspot
Data Completeness Rate	AVG([Is_Complete_Record]) * 100	Measures data governance quality

In Tableau, the KPI tiles should be built using calculated fields and FIXED LOD expressions where required so that executive metrics remain stable and consistent across dashboard interactions.

SECTION 8 — PIVOT TABLE ANALYSIS

The project uses 10 pivot tables to convert ticket-level data into decision-ready analytical summaries. These pivot tables support Tableau sheets, statistical tests, and recommendations.

Pivot 1 — Borough × Violation Type

This matrix shows how violation types differ across boroughs. It is the foundation for borough-specific enforcement policy and is used directly for the chi-square test.

Business Insight: If each borough has a different violation mix, a uniform city-wide enforcement plan is inefficient.

Pivot 2 — Borough × Month

This pivot identifies seasonal enforcement peaks by borough.

Business Insight: If different boroughs peak in different months, staffing should be seasonally adjusted rather than fixed throughout the year.

Pivot 3 — Violation Type × Fine Category

This table separates high-frequency low-value violations from low-frequency high-value violations.

Business Insight: Revenue-focused enforcement and safety-focused enforcement may require different priorities.

Pivot 4 — Day of Week × Violation Type

This pivot identifies when specific violations cluster.

Business Insight: Street cleaning violations may peak on weekdays, while other violations may remain stable across the week.

Pivot 5 — State Group × Borough

This pivot maps where out-of-state vehicles receive tickets.

Business Insight: NJ and CT plate concentration near transit corridors can justify interstate DMV data-sharing for fine recovery.

Pivot 6 — Vehicle Type Group × Violation Severity

This pivot shows whether commercial, taxi, passenger, or transit vehicles produce more safety-critical violations.

Business Insight: If commercial vehicles dominate safety-critical violations, the city should expand loading zones instead of only increasing penalties.

Pivot 7 — Offender Tier × Borough

This pivot identifies where repeat offenders are concentrated.

Business Insight: The city can launch repeat offender escalation programs in the boroughs with the highest 6+ ticket tier concentration.

Pivot 8 — Street × Violation Type

This table identifies top street-level hotspots and their dominant violation types.

Business Insight: A street dominated by street cleaning violations needs signage improvements, while a street dominated by fire hydrant violations needs stricter safety enforcement.

Pivot 9 — Vehicle Age Group × Violation Description

This pivot connects vehicle age to violation behavior.

Business Insight: Older vehicles may require targeted DMV renewal or inspection awareness campaigns.

Pivot 10 — Month × Borough Average Fine Calendar

This pivot combines time, geography, and revenue intensity.

Business Insight: Boroughs with high average fines in specific months should receive targeted seasonal enforcement.

SECTION 9 — EXPLORATORY DATA ANALYSIS

9.1 EDA Objective

The EDA process was designed to move beyond simple counts and reveal patterns in safety, revenue, compliance, timing, geography, and offender concentration.

The project includes 10 EDA analyses: null heatmap, top violations, fine distribution, monthly trend, day-month heatmap, borough revenue bar, normalised borough heatmap, Pareto repeat offender chart, correlation matrix, and avoidable vs safety split.

9.2 Major EDA Findings

E1 — Null Heatmap

The null heatmap identifies whether missing values are random or systematic.

Interpretation: If vehicle color and vehicle make are missing together, it may indicate officer-level data entry issues rather than random missingness.

E2 — Top 10 Violations

A horizontal bar chart colored by violation severity separates administrative, standard, and safety-critical violations.

Interpretation: If most violations are administrative, the city may be optimizing for revenue rather than public safety.

E3 — Fine Distribution

The histogram and boxplot show whether fines cluster around standard amounts such as $65 and $115.

Interpretation: Fine structure is likely violation-code driven rather than context-driven.

E4 — Monthly Trend

A dual-axis chart compares ticket count and fine revenue over months.

Interpretation: Months with high ticket count but lower revenue indicate dominance of low-fine violations.

E5 — Day × Month Heatmap

This heatmap shows enforcement rhythm across weekdays and months.

Interpretation: Consistent weekday peaks may reflect street-cleaning schedules or deployment patterns.

E6 — Borough Revenue Bar

This chart compares total borough revenue while coloring by average fine.

Interpretation: A borough with fewer tickets but higher average fine may be more enforcement-intensive in safety-critical violations.

E7 — Normalized Borough Heatmap

This chart shows violation type as a percentage within each borough.

Interpretation: Normalization prevents Manhattan or high-volume boroughs from visually dominating the analysis.

E8 — Repeat Offender Pareto Chart

This chart shows cumulative fine revenue by plate.

Interpretation: If a small percentage of plates generate most revenue, repeat offender targeting becomes the highest-impact policy lever.

E9 — Correlation Matrix

The matrix checks relationships between numeric variables such as fine amount, vehicle age, repeat count, weekend flag, and out-of-state status.

Interpretation: Low correlation with fine amount suggests fines are structurally tied to violation code rather than contextual conditions.

E10 — Avoidable vs Safety Split

This stacked bar compares avoidable and safety/intentional violation revenue by borough.

Interpretation: Boroughs with high avoidable violation share are ideal candidates for signage and communication pilots.

SECTION 10 — STATISTICAL ANALYSIS

10.1 Purpose

Statistical testing was used to validate whether observed dashboard patterns were meaningful or likely due to random variation. The project includes six tests: t-test, chi-square, ANOVA with Tukey HSD, Pearson correlation, time-series decomposition, and linear regression.

Test 1 — Independent T-Test

Question: Do NY and out-of-state plates pay different average fines?
Null Hypothesis: Average fines for NY and out-of-state plates are equal.

Business Interpretation:
If significant, out-of-state vehicles may be associated with higher-value violations. If not significant, fine amounts are likely determined by violation type, while the real policy issue remains collection risk.

Test 2 — Chi-Square Test

Question: Is violation type independent of borough?
Null Hypothesis: Violation distribution is identical across all boroughs.

Business Interpretation:
If the test rejects the null hypothesis, borough-specific enforcement strategies are statistically justified.

Test 3 — One-Way ANOVA + Tukey HSD

Question: Do boroughs have different average fine amounts?
Null Hypothesis: All boroughs have equal average fines.

Business Interpretation:
If significant, enforcement value differs by borough. Tukey HSD identifies which borough pairs differ the most.

Test 4 — Pearson Correlation

Question: Is vehicle age related to ticket frequency per plate?
Null Hypothesis: There is no relationship between vehicle age and ticket frequency.

Business Interpretation:
A positive correlation may suggest that older vehicles are more associated with repeat ticketing or registration-related violations.

Test 5 — Time Series Decomposition

Question: Are ticket volumes seasonal?

Business Interpretation:
A strong seasonal component indicates that enforcement planning should follow a seasonal calendar rather than a fixed monthly deployment model.

Test 6 — Linear Regression

Question: Can day, month, and borough predict fine amount?

Expected Interpretation:
A low R^2 is still meaningful because it suggests fine amounts are mostly fixed by violation code rather than contextual variables. This supports the conclusion that the penalty system is rigid and not adaptive.

SECTION 11 — TABLEAU DASHBOARD DESIGN

The Tableau system contains 12 sheets and 4 dashboards. The sheets include KPI tiles, monthly trend, borough revenue bar, violation treemap, normalized heatmap, day-of-week bar, street stacked bar, vehicle make bubble chart, repeat offender Pareto, out-of-state views, severity donut, and vehicle age histogram.

Dashboard 1 — Executive Overview: “The City Picture”

Audience: Director-level officials and senior decision makers
Purpose: Communicate the overall scale, revenue, severity, and city-wide enforcement picture in under 90 seconds.

Sheets Used

- S1: KPI Summary Tiles
- S2: Monthly Trend Dual-Axis Chart
- S3: Borough Revenue Bar
- S11: Violation Severity Donut
- S5: Borough × Violation Heatmap

Layout

Row	Components
Row 1	Six KPI tiles in a 3×2 grid
Row 2	Monthly trend chart on left, severity donut on right
Row 3	Borough revenue bar on left, normalized borough heatmap on right

Filters

- Date range slider

- Borough multi-select
- Plate type dropdown

Dashboard Story

This dashboard answers:
How large is the enforcement problem, where is the money concentrated, and is enforcement focused on safety or revenue?

Dashboard 2 — Operational Analysis: “When and What”

Audience: Operations managers and precinct commanders
Purpose: Help enforcement teams decide when officers should be deployed and which violations should be prioritized.

Sheets Used

- S4: Violation Treemap
- S5: Borough × Violation Heatmap
- S6: Day-of-Week Bar
- Avoidable vs Safety Stacked Bar

Layout

Row	Components
Row 1	Violation treemap on left, day-of-week bar on right
Row 2	Normalised borough × violation heatmap full width
Row 3	Avoidable vs safety stacked bar full width

Filters

- Month selector
- Violation severity toggle
- Weekend toggle

Dashboard Story

This dashboard answers:
When should officers be deployed, and which violation types should receive operational priority?

Dashboard 3 — Vehicle Intelligence: “Who Is Being Ticketed”

Audience: Policy analysts and urban planners
Purpose: Understand vehicle type, vehicle make, street hotspots, and vehicle age patterns.

Sheets Used

- S7: Top 10 Streets Stacked Bar
- S8: Vehicle Make Bubble Chart
- S10: State Group Donut and State × Borough Table
- S12: Vehicle Age Histogram

Layout

Row	Components
Row 1	Top streets stacked bar on left, vehicle age histogram on right
Row 2	Vehicle make bubble chart on left, body type donut on right
Row 3	State group donut and state × borough text table

Filters

- Vehicle type group
- Registration state
- Borough

Dashboard Story

This dashboard answers:
Which vehicle groups, streets, and states contribute most to parking violations and safety risk?

Dashboard 4 — Offender Intelligence: "The Repeat Problem"

Audience: Policy decision makers, compliance teams, and legal teams
Purpose: Identify repeat offender concentration and escalation opportunities.

Sheets Used

- S9: Repeat Offender Pareto Chart
- S10: Out-of-State Analysis
- Repeat Offender × Borough Bar
- KPI Tiles

Layout

Row	Components
Row 1	Repeat offender %, revenue from 6+ tier, unique plates
Row 2	Pareto chart full width
Row 3	Repeat offender × borough bar full width

Filters

- Offender tier
- Borough
- Registration state

Dashboard Story

This dashboard answers:
Are a small number of vehicles creating a disproportionate share of violations and fine revenue?

SECTION 12 — INSIGHTS SUMMARY

Insight 1 — Ticket Count Does Not Equal Enforcement Quality

High ticket volume does not automatically mean better enforcement. The average fine and severity mix must also be considered.

Insight 2 — Borough-Level Differences Are Strategically Important

Different boroughs may have different violation profiles. This means enforcement policies should be localized rather than uniform.

Insight 3 — Repeat Offenders Are a High-Value Target

A small group of repeat offenders may generate a disproportionate share of revenue and enforcement workload.

Insight 4 — Avoidable Violations Create a Prevention Opportunity

A large avoidable violation share suggests that better signage, reminders, or digital parking guidance could reduce unnecessary violations.

Insight 5 — Commercial Vehicle Violations Need Infrastructure Solutions

If commercial vehicles are concentrated in safety-critical violations, the city should consider loading-zone expansion and curb management reforms.

Insight 6 — Out-of-State Tickets Represent Collection Risk

Out-of-state vehicles may create revenue leakage because fine recovery is more difficult across jurisdictions.

Insight 7 — Street-Level Hotspots Need Targeted Interventions

Top streets should not all receive the same solution. Each street's dominant violation type should determine the intervention.

Insight 8 — Fine Amounts Are Structurally Driven

Regression analysis is expected to show that fine amount is mainly determined by violation type, not by contextual factors like month or borough.

Insight 9 — Weekend Enforcement May Be Underutilized

Weekend enforcement percentage helps determine whether weekend patrols are underfunded or intentionally lower.

Insight 10 — Data Completeness Is a Governance Issue

Missing or inconsistent vehicle fields may indicate training issues, officer-level reporting gaps, or system design problems.

SECTION 13 — RECOMMENDATIONS

Recommendation 1 — Deploy Violation-Calibrated Patrol Schedules

Priority: High
Category: Operations

Insight

Day-of-week and borough-month patterns show that violations are not evenly distributed across time and geography.

Action

Use day × violation and borough × month matrices to create a data-driven patrol calendar.

Expected Impact

A 15–20% improvement in ticket yield per officer-hour is expected if deployment is aligned with high-violation windows.

Implementation Pathway

- Build monthly deployment calendar
- Prioritize high-volume borough-day combinations
- Review staffing efficiency every quarter
- Update patrol plans based on Tableau dashboard trends

Recommendation 2 — Launch Repeat Offender Escalation Program

Priority: High
Category: Compliance and Policy

Insight

The Pareto analysis can identify whether a small share of plates generates a major share of fine revenue.

Action

Create a graduated escalation system:

- Warning letter at 3 tickets
- Compliance notice at 5 tickets
- Booting eligibility at 7 tickets
- DMV renewal flag for chronic unpaid offenders

Expected Impact

A 25% reduction in repeat-visit enforcement workload is possible if chronic violators are removed from the repeat cycle.

Implementation Pathway

- Use offender tier table
- Flag 6+ tier plates
- Launch pilot in highest repeat-offender borough
- Monitor repeat rate reduction monthly

Recommendation 3 — Install Dynamic Digital Parking Signs on High-Avoidable Streets

Priority: Medium
Category: Infrastructure

Insight

Avoidable violations represent preventable behavior caused by confusion, unclear signage, or missed timing rules.

Action

Install dynamic LED parking signs on the top five high-avoidable-violation streets.

Expected Impact

A 10–15% reduction in avoidable citations on target streets.

Implementation Pathway

- Identify top streets from street summary
- Select pilot streets by avoidable violation share
- Install digital signs
- Compare violation rates before and after installation

Recommendation 4 — Expand Loading Zones on Commercial Safety Hotspots

Priority: Medium
Category: Commercial Policy and Urban Planning

Insight

Commercial and taxi vehicles may be overrepresented in safety-critical violations.

Action

Expand time-limited loading zones on streets where commercial vehicles frequently receive safety-critical violations.

Expected Impact

A 20% reduction in double-parking and blocking violations on target streets.

Implementation Pathway

- Use Vehicle Type × Violation Severity pivot
- Identify top commercial safety streets
- Add loading zone pilots
- Monitor fire hydrant, bus stop, and blocking-intersection violations

Recommendation 5 — Improve Out-of-State Fine Recovery Through DMV Data Sharing

Priority: Lower to Medium
Category: Revenue Recovery and Cross-Jurisdiction Policy

Insight

Out-of-state tickets represent potential revenue leakage, especially when concentrated among nearby states such as NJ and CT.

Action

Create or strengthen DMV data-sharing agreements for unpaid parking fines.

Expected Impact

Potential recovery of $150K–$300K annually from high-risk out-of-state tickets, depending on final dashboard values.

Implementation Pathway

- Identify top out-of-state groups
- Prioritize NJ and CT if they dominate out-of-state volume
- Develop inter-agency MOU
- Add renewal-hold mechanism for unpaid fines

SECTION 14 — IMPACT ESTIMATION

Recommendation	Estimated Impact	Decision Area
Patrol schedule optimization	15–20% improvement in officer-hour yield	Operations
Repeat offender escalation	25% reduction in repeat enforcement workload	Compliance
Dynamic parking signs	10–15% reduction in avoidable citations	Infrastructure
Loading zone expansion	20% reduction in commercial safety violations	Urban planning
DMV data-sharing	$150K–$300K annual recovery potential	Revenue recovery

These impact estimates should be treated as planning estimates until validated with final Tableau dashboard values and historical before-after comparisons.

SECTION 15 — LIMITATIONS

15.1 Data Limitations

- The dataset captures issued tickets, not all actual violations.
- It does not show whether tickets were paid, disputed, or dismissed.
- Plate IDs may be anonymized or inconsistent.
- Some vehicle fields may be missing or incorrectly entered.
- The dataset may not include officer deployment data, which limits productivity analysis.

15.2 Analytical Limitations

- Correlation does not imply causation.
- Dashboard patterns show association, not direct policy impact.
- Estimated recommendation impacts require real-world pilot validation.
- Fine amounts are structurally tied to violation codes, limiting predictive modeling value.

15.3 Tableau Limitations

- Aggregated exports improve performance but reduce row-level flexibility.
- Some KPIs require FIXED LOD expressions to avoid filter distortion.
- Very detailed street-level charts may need filtering to avoid clutter.

SECTION 16 — FUTURE SCOPE

16.1 Predictive Enforcement Model

Build a machine learning model to predict high-risk borough-street-day combinations.

16.2 Payment and Collection Analysis

Add payment status, late fees, and dispute outcomes to understand true revenue realization.

16.3 Officer Productivity Analysis

Integrate officer deployment logs to calculate tickets per officer-hour.

16.4 Real-Time Parking Risk Dashboard

Create a live dashboard showing current high-risk streets based on historical patterns.

16.5 Equity and Fairness Analysis

Analyse whether enforcement disproportionately affects certain neighborhoods or vehicle groups.

16.6 Geospatial Mapping

Add latitude and longitude to create hotspot maps in Tableau or GIS tools.

SECTION 17 — CONCLUSION

This project transforms NYC parking ticket data into a complete decision-support system for enforcement strategy, revenue intelligence, safety prioritization, and policy planning. The analysis shows that parking violations should not be understood only through total ticket volume. A more meaningful view combines revenue, violation severity, borough differences, repeat offender behavior, vehicle type, avoidable violations, and data quality.

The four Tableau dashboards serve different audiences: executives, operations teams, policy analysts, and compliance decision makers. The strongest policy direction comes from three findings: boroughs need localized enforcement strategies, repeat offenders require escalation, and avoidable violations should be reduced through infrastructure and communication rather than enforcement alone.

The project demonstrates how data visualization and analytics can convert municipal ticket records into actionable governance intelligence.

SECTION 18 — APPENDIX

18.1 Data Dictionary

Variable	Definition
Summons_Number	Unique ticket identifier
Issue_Date	Date when ticket was issued
Violation_Code	Code representing violation type
Violation_Description	Text description of violation
Violation_County	Borough/county of violation
Fine_Amount	Fine value
Plate_ID	Vehicle plate identifier
Registration_State	State where vehicle is registered
Vehicle_Make	Manufacturer of vehicle
Vehicle_Body_Type	Vehicle body category
Vehicle_Color	Vehicle color
Street_Name	Street where ticket was issued
Month_Name	Month extracted from issue date
Day_of_Week	Day name extracted from issue date
Is_Weekend	Weekend flag
Is_OutOfState	Out-of-state registration flag
Vehicle_Age	Derived vehicle age
Vehicle_Type_Group	Grouped vehicle type
Violation_Severity	Safety-Critical, Standard, Administrative
Is_Avoidable	Whether violation is preventable through better information
Is_Repeat_Offender	Whether plate received repeat tickets
Offender_Tier	Ticket frequency tier
Is_Complete_Record	Completeness indicator

18.2 Tableau Sheet Summary

Sheet	Chart	Dashboard
S1	KPI Tiles	DB1
S2	Monthly Trend Dual-Axis	DB1
S3	Borough Revenue Bar	DB1
S4	Violation Treemap	DB2
S5	Borough × Violation Heatmap	DB1, DB2
S6	Day-of-Week Bar	DB2
S7	Top Streets Stacked Bar	DB3
S8	Vehicle Make Bubble Chart	DB3
S9	Repeat Offender Pareto	DB4
S10	Out-of-State Donut + Table	DB3, DB4
S11	Violation Severity Donut	DB1
S12	Vehicle Age Histogram	DB3

18.3 Cleaning Summary

Step	Action	Purpose
Missing values	Audited and cleaned null-heavy fields	Improve data quality
Dates	Standardised date formats	Enable time-series analysis
Fine amount	Corrected invalid values	Ensure accurate revenue metrics
Vehicle year	Removed invalid years	Enable vehicle age analysis
Colors	Mapped abbreviations	Standardise vehicle attributes
Boroughs	Standardised names	Enable correct grouping
Makes	Consolidated inconsistent names	Improve vehicle analysis
States	Validated registration state	Enable out-of-state analysis
Streets	Standardised street names	Enable hotspot analysis
Plates	Created repeat offender flags	Enable offender intelligence
Duplicates	Removed duplicate summons	Prevent double counting
Validation	Applied final assertions	Ensure analysis-ready output

18.4 Contribution Matrix

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Kavya Katal				✓		✓	✓
Ayush Tiwari	✓	✓	✓				
Vaibhav Rajput			✓			✓	✓
Abuzar Haideri			✓	✓	✓		

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